6	5	4	3	2	1

Note 3-1:	C104
MT6735	NC
MT6735M	NC
MT6735P	NC
MT6737	NC
MT6737M	NC
MT6737T	NC
MT6753	47uF
MT6753T	47uF

FSOURCE_P(EFUSE)
(1)FSOURCE_P EFUSE power(VEFUSE) should be
only for EFUSE usage(not share with other application)
(2)W/I EFUSE program, VEFUSE need 1uF bypass cap
(pls refer to ¡ §LDO output voltage/current table¡ ¨)
(3)W/O EFUSE program, VEFUSE bypass cap should be NC.

DRAWN: Longbaicheng	DATED: 2019-01-13	COMPANY: TRANSSION	
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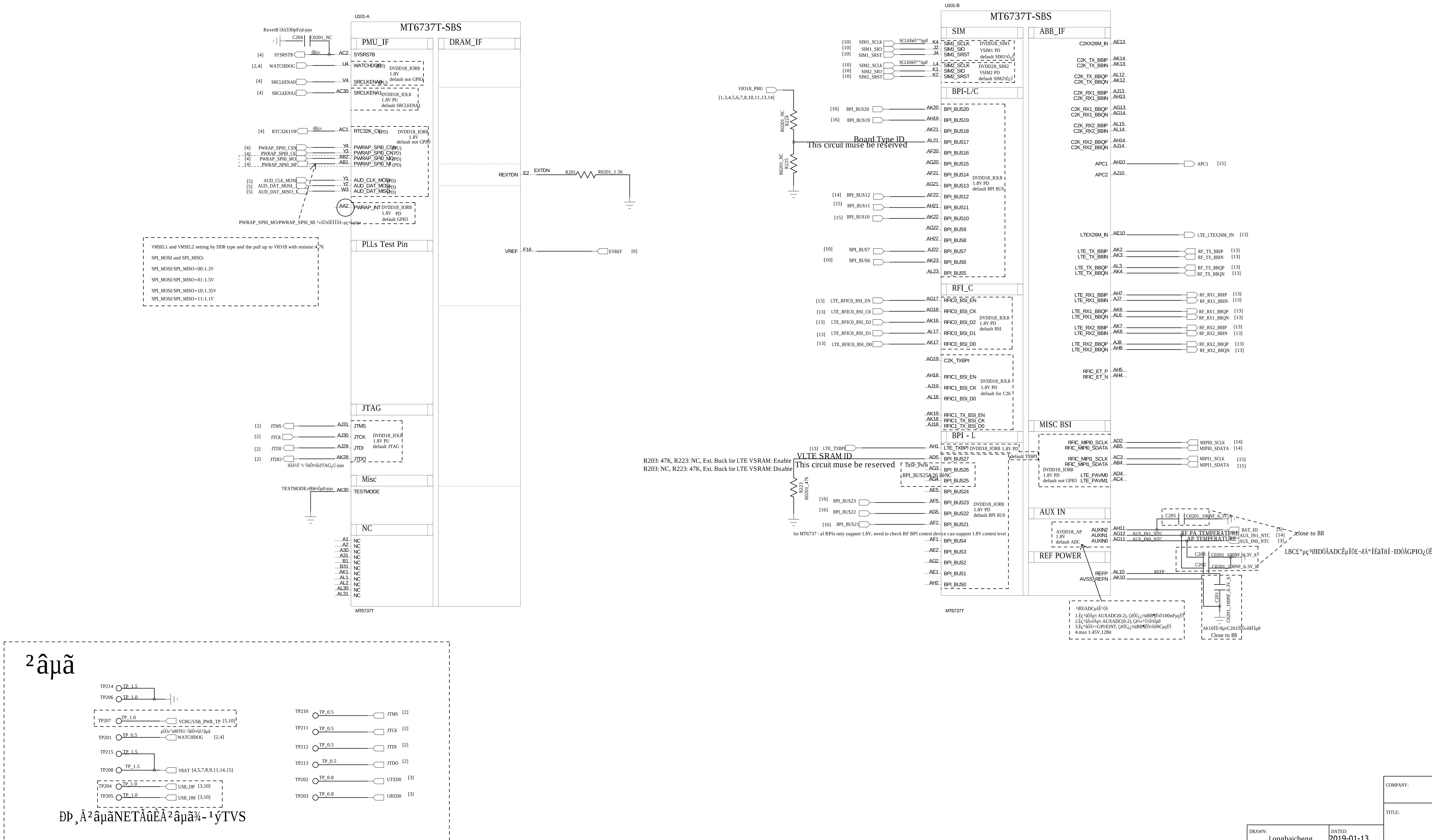

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TITLE:	INF-NOTE4 LITE_MT6737T_Schematic		
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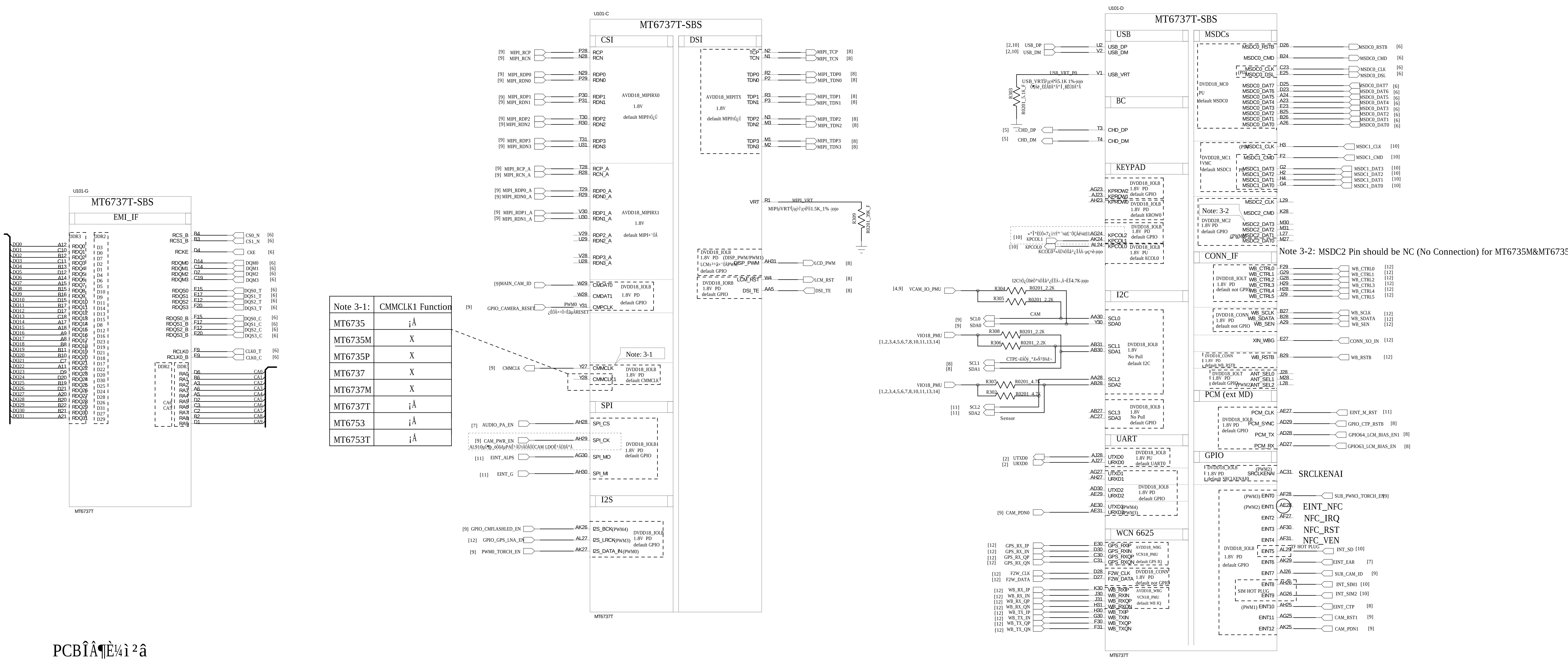
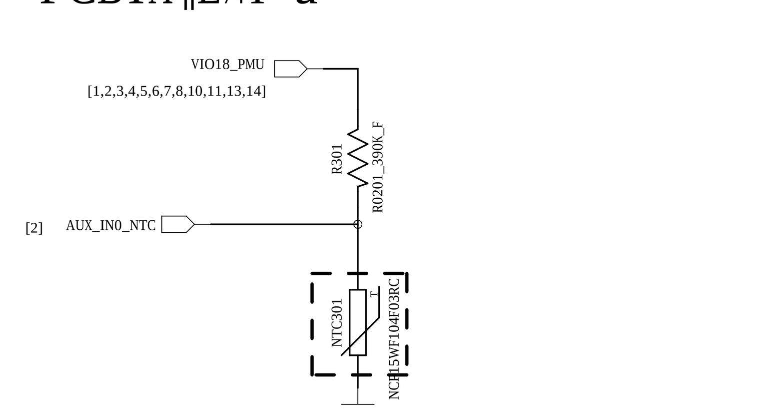


Note 3-2: MSDC2 Pin should be NC (No Connection) for MT6735M&MT6735P

Note 3-1:	CMMCLK1 Function
MT6735	¡Â
MT6735M	X
MT6735P	X
MT6737	X
MT6737M	X
MT6737T	¡Â
MT6753	¡Â
MT6753T	¡Â

PCBÎÂ¶È¼ì ²â



NTC¾àAP 5-8mm£¬Í·Ê±±Ö¿ª£ªû·¢ÊÈ£«¾þ£¯PMIC,Charger,WIFI¡£Ê£15mmÒÔÊÏ

SET	Controller	Ref.Design	Note
I2C0	AP	Rear Camera w AF Front Camera w/o AF	1.Open drain Type 2.PU to Camera's power supply
I2C1	AP	CTP/SW Charger LCM gate driver	1.Open drain Type
I2C2	AP	MEMS sensor/NFC Flashlight Driver MHL/Front Camera w AF	1.Open drain Type
I2C3	AP	reserved for future use	1.Open drain Type

Set	Function	Power Domain	Note
MSDC0	eMMC	DVDD18_MC0	1.Dedicated pin for eMMC 2.Independent power domain
MSDC1	T-Card	DVDD28_MC1	1.Dedicated pin for T-Card 2.Independent power domain
MSDC2	MT6630	DVDD28_MC2	1.1.8V for SDIO3.0

COMPANY: TRANSSION			
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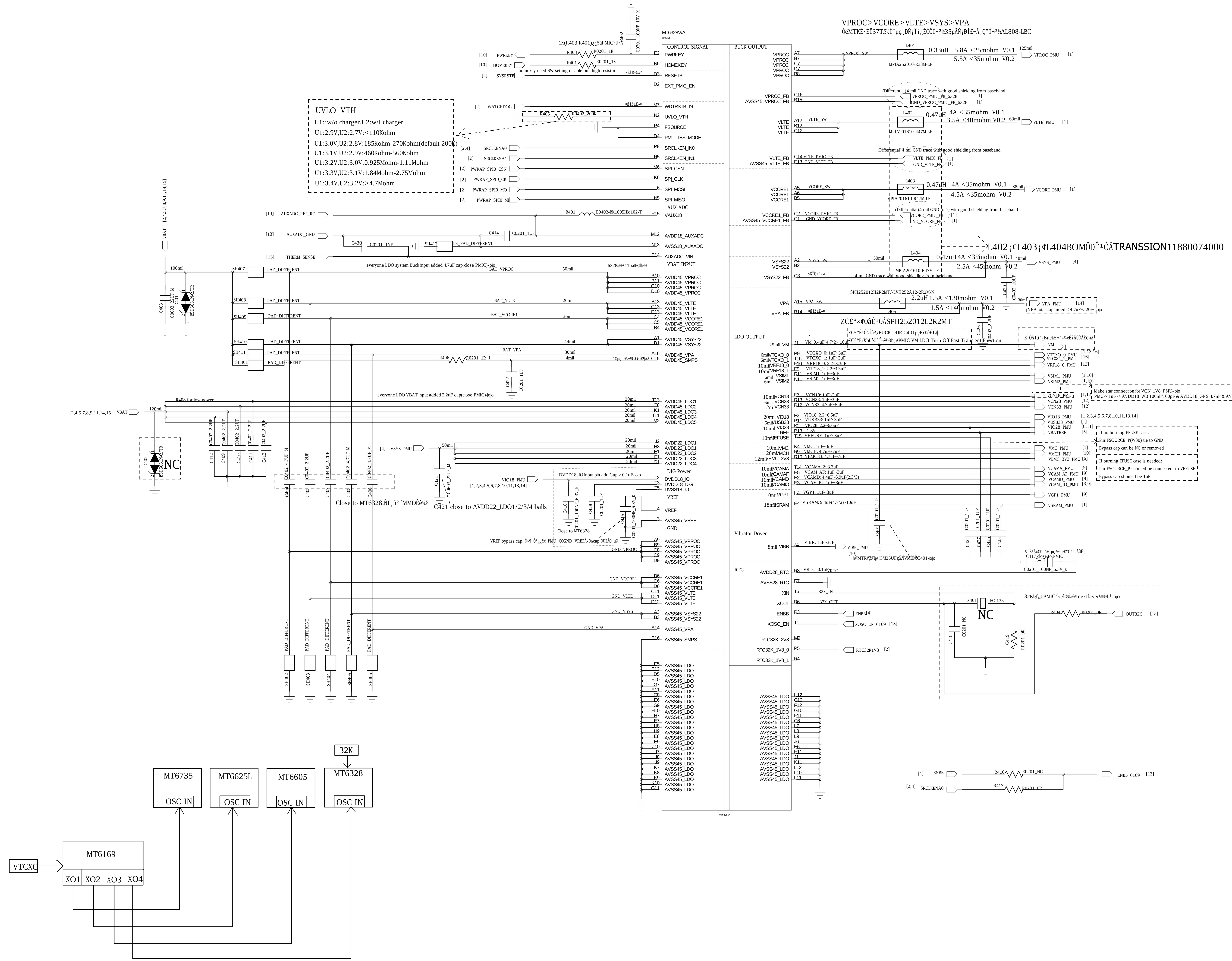

MT6735 Core Powers	PMIC Power source	Buck/LDO	Min. Voltage	Max. Voltage
VPROC(DVDD_CPU)	MT6328's VPROC	Buck	0.85	1.265
VCORE(DVDD_CORE)	MT6328's VCORE	Buck	0.85	1.265
VLTE(DVDD_LTE)	MT6328's VLTE	Buck	0.85	1.155
VM(DVDD_EMI)	MT6328's VM	LDO	1.14	1.300
VSRAM(DVDD_SRAM)	MT6328's VSRAM	LDO	0.85	1.310
LTE_VSRAM_EXT(VLTE_SRAM)	External Buck:LTE_VSRAM_EXT	Buck	0.85	1.150

BUCK	Output Voltage(V)	Output Current(mA)	Supply
VPROC	1.15V[0.6V-1.31V]6.25mV	5000	CPU Processor
VCORE	1.15V[0.6V-1.31V]6.25mV	3500	GPU+Soc
VLTE	1.05V[0.6V-1.31V]6.25mV	2800	LTE Core
VSYS	2V	1900	System Power
VPA	3.4V(0.5V-3.4V)	600	MT6169 PA

LDO	Output Voltage(V)	Output Current(mA)	Supply
VTCXO0	2.8	40	MT6169 TCXO
VTCXO1	2.8	40	MT6158 TCXO
VCN33	3.3	350	MT6625L 3.3V
VCN28	2.8	40	MT6625L 2.8V
VAUX18	1.8V	40	AUXADC 1.8V
VAUD28	2.8(1.5/1.8/2.5/2.8)	40	Audio Power
VCAMA	2.8(1.2/1.3/1.5/1.8/2.0/2.8/3.0/3.3)	200	Camera Analog power
VMCH	2.9(2.9/3.0/3.3)	800	SD 3.0
VCAM_AF	2.8(1.2/1.3/1.5/1.8/2.0/2.8/3.0/3.3)	200	Camera AF power
VGP1	2.8(1.2/1.3/1.5/1.8/2.5/2.8/3.0/3.3)	200	CTP
VIO28	2.8	200	Sensor,IR LED
VIBR	2.8(1.2/1.3/1.5/1.8/2.0/2.8/3.0/3.3)	100	MOTO
VMC	2.9(1.8/2.9/3.0/3.3)	200	SD Card 3.0
VEFUSE	1.8(1.8/1.9/2.0/2.1/2.2)	200	Efuse
VEMC33	3.3(3.0/3.3)	400	EMMC 5.0 power
VUSB	3.3	20	USB 2.0
VSIM1	1.8(1.7/1.8/1.86/2.76/3.0/3.1)	50	SIM CARD 1
VSIM2	1.8(1.7/1.8/1.86/2.76/3.0/3.1)	50	SIM CARD 2
VRTC	2.8	2	RTC
DVDD18_DIG	1.8	20	
VCAMD	(0.9/1.0/1.1/1.22/1.3/1.5)	500	Camera Digital power

¾§Ìå·½°¸ÌùÆ¬ÅäÖÃ±í1/3 (Page4¡¢13¡¢16£©

	MT6328 AuxADC			32K Crystal Component			
	C430	C414	B401	R404	C418	C419	X401
VCTXO+32K Exist	0R	1uF	0R	NC	22pF	22pF	32K
TSX+32K Exist	1nF	1uF	Bead	NC	22pF	22pF	32K
TSX+32K Less	1nF	1uF	Bead	0R	NC	0R	NC

COMPANY: TRANSSION

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QUALITY CONTROL: <QC By>	DATED: <QC Date>			
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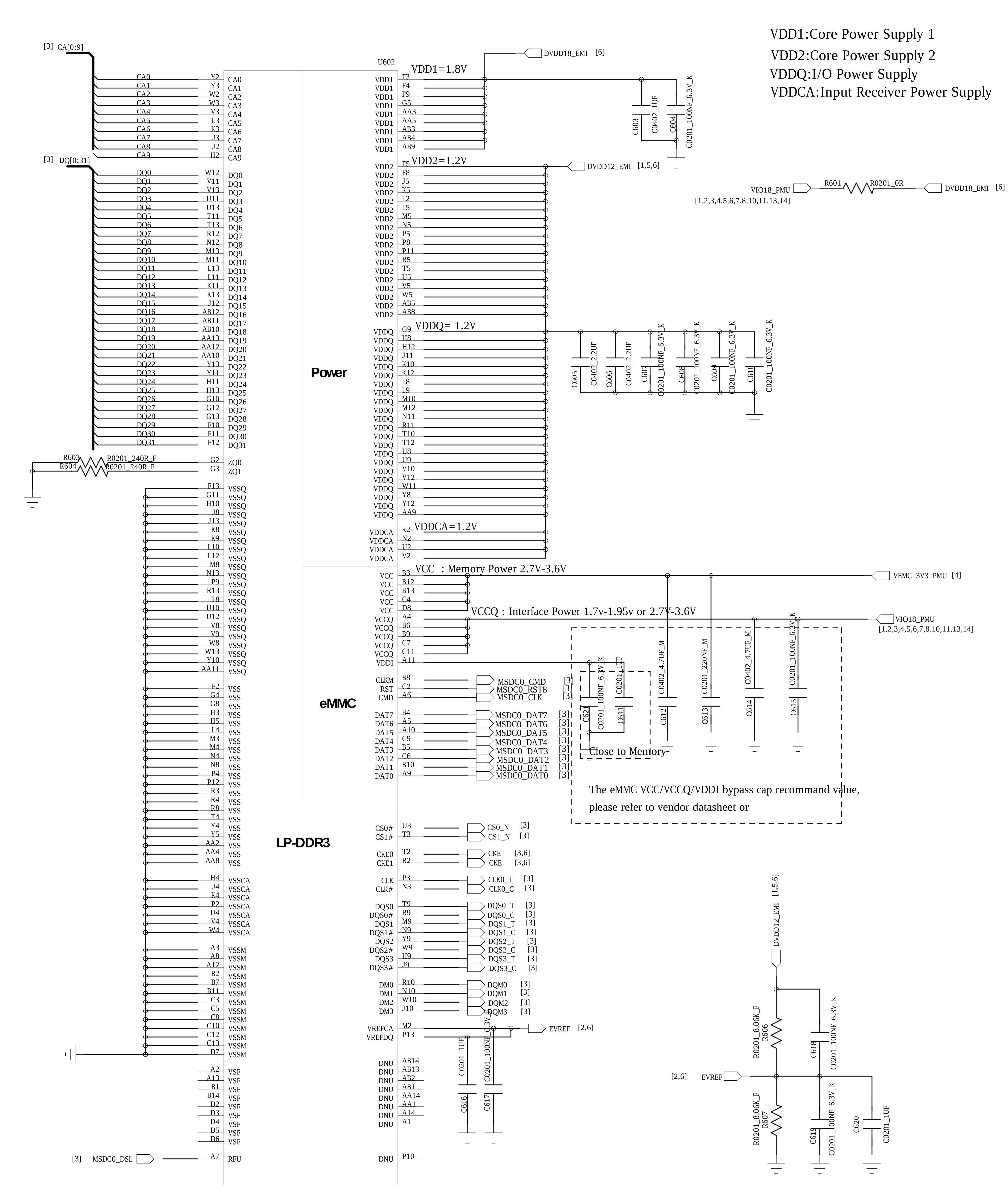
eMMC+LPDDR3



LPDDR3 FlashÐÅÏ¢»ã×Ü

221 Ball, 0.5mm pitch

ÈÝÁ¿	ÐÍºÅ	³§¼Ò	³ß´ç	ÈÏÖ¤Çé¿ö
16GB+2GB	KMQE10013M-B318007	SAMSUNG	1.0	ÈÏÖ¤OK
	H9TQ17ABJTACUR-KUM	HYNIX	1.0	Î´ÈÏÖ¤
	TYE0HH231659RA	TOSHIBA	1.0	ÈÏÖ¤OK

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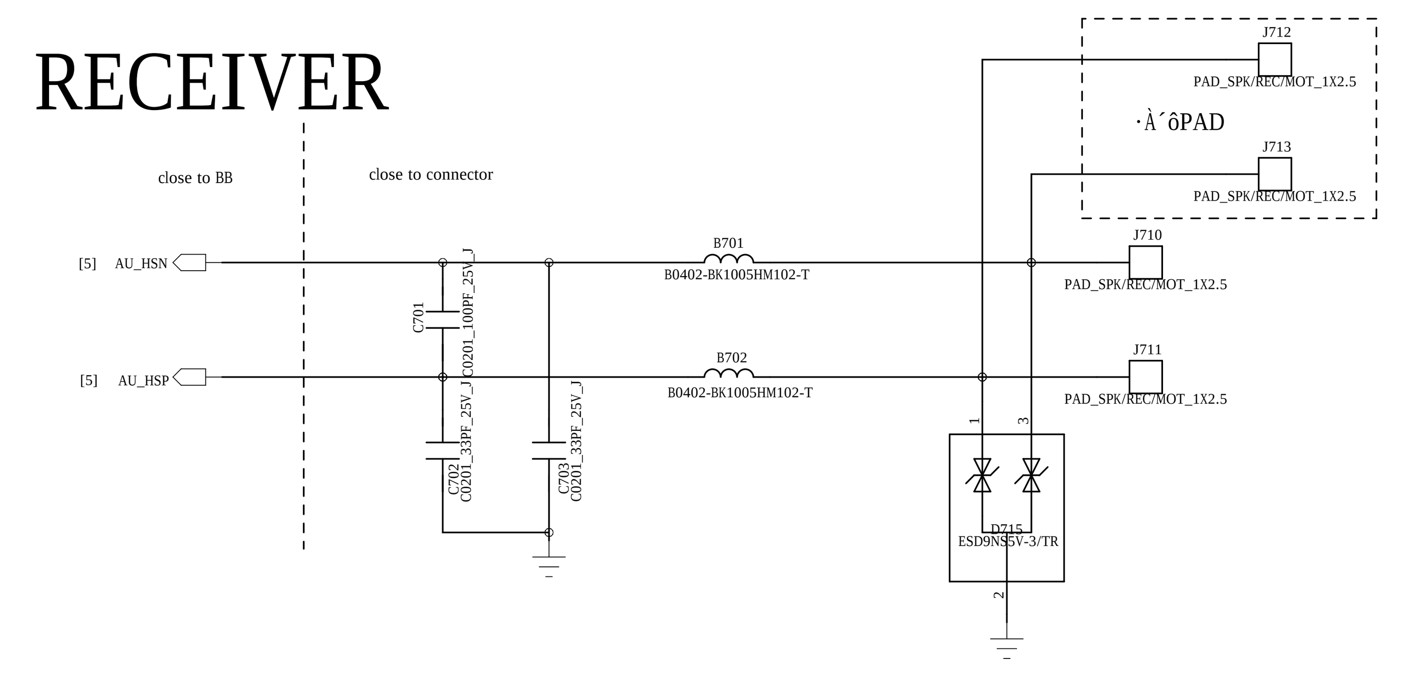
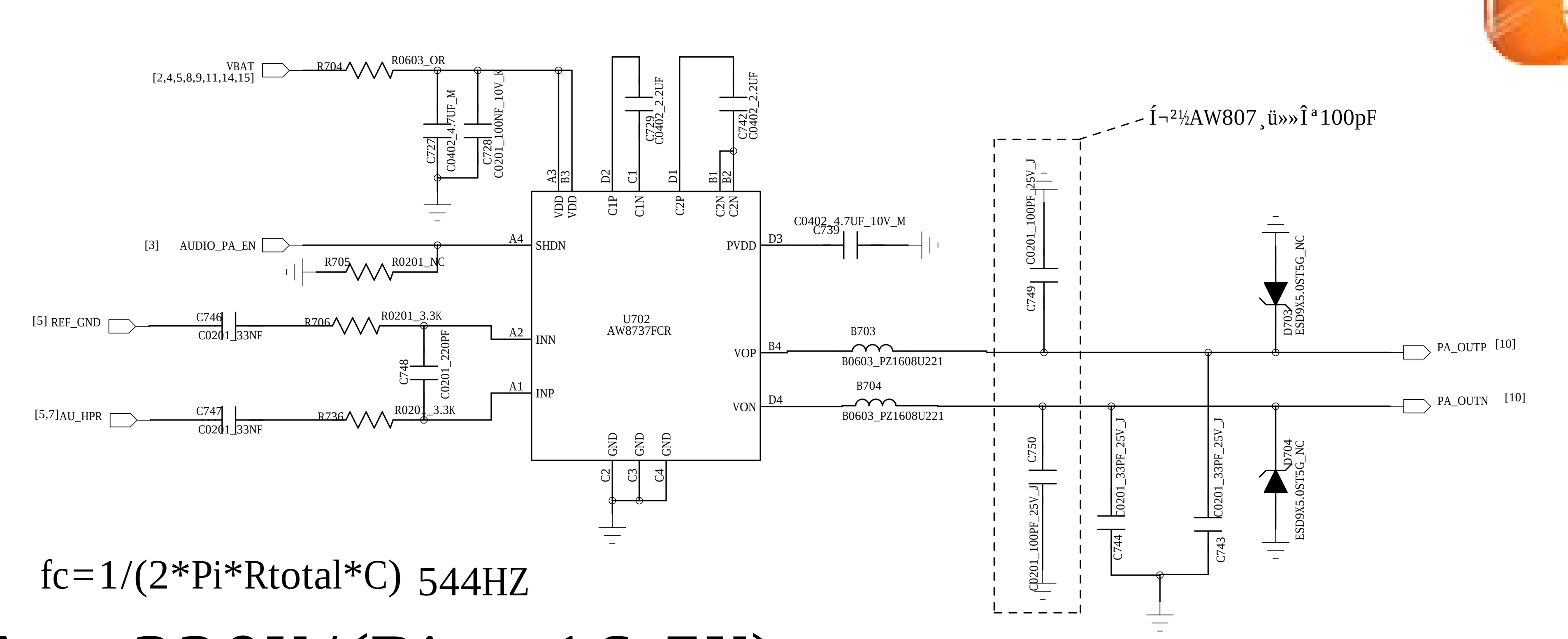
SCALE: <Scale>	SHEET: 6 OF 17			

RECEIVER



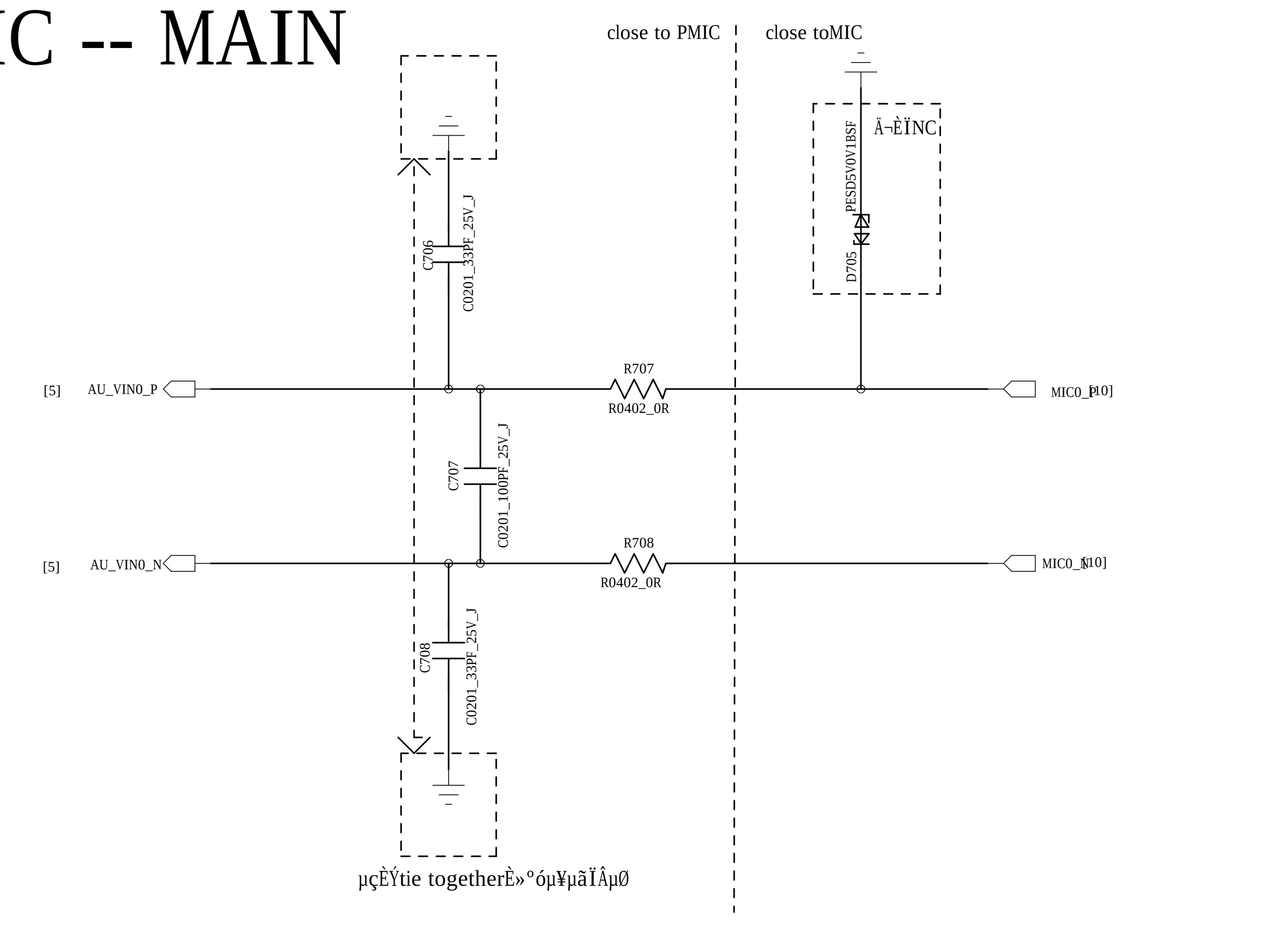
AUDIO PA



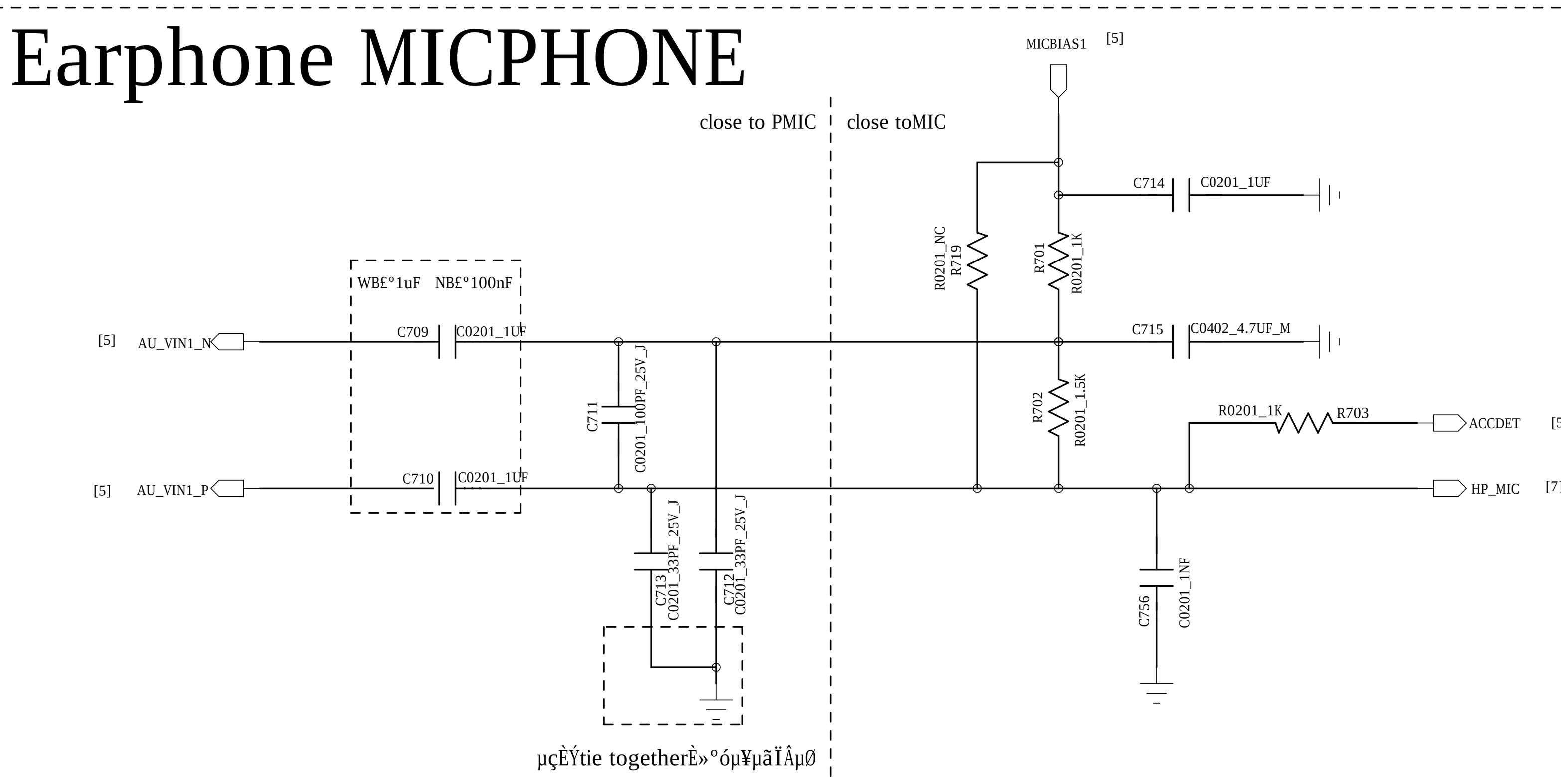
fc=1/(2*Pi*Rtotal*C) 544HZ

Av=320K/(Rin+16.5K)

MIC -- MAIN

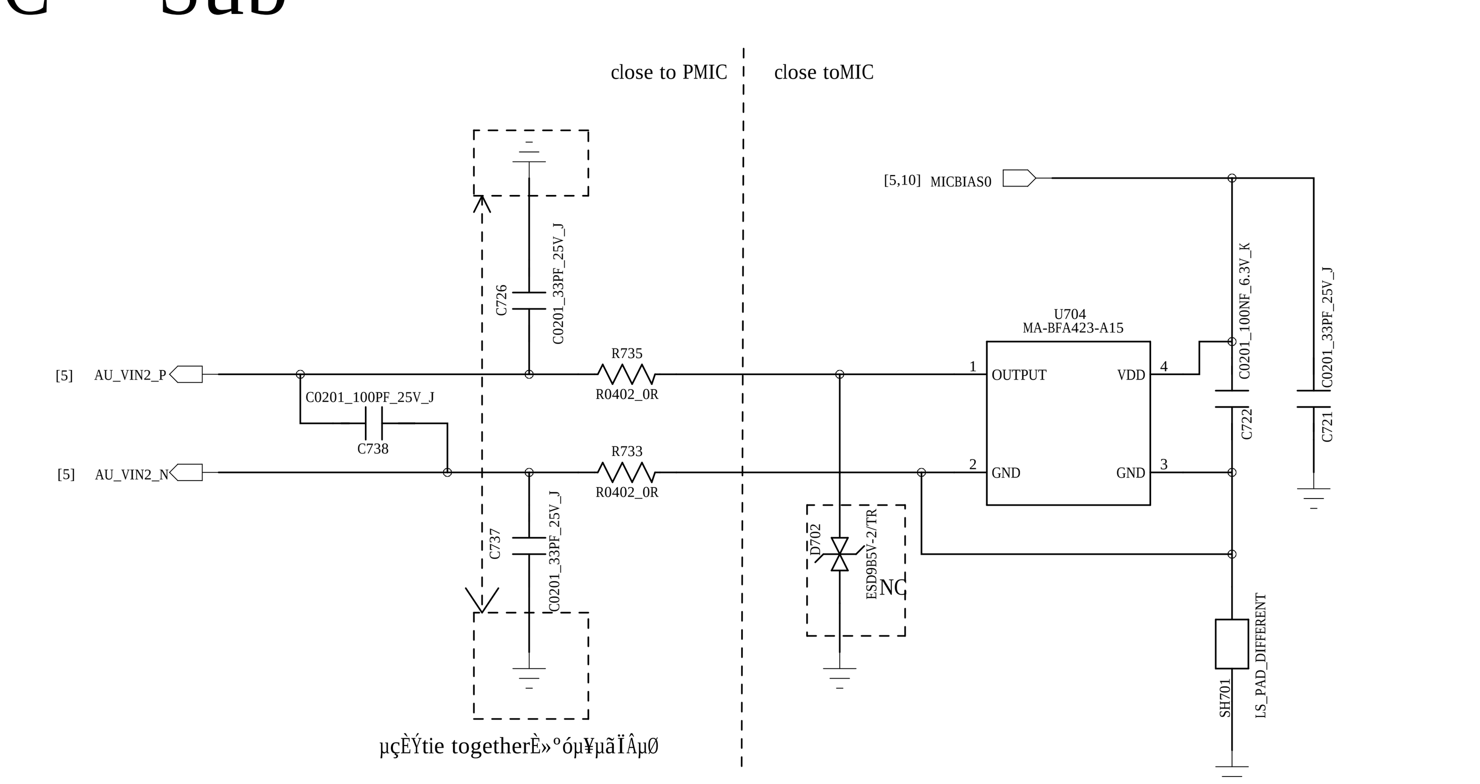


Earphone MICPHONE

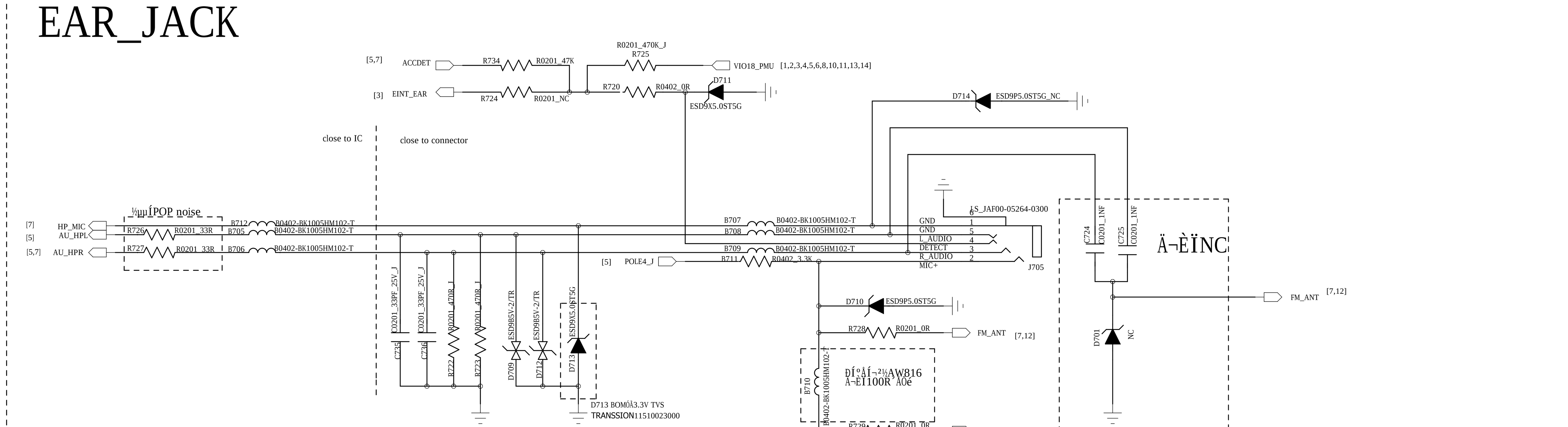


Mic mode setting									
Mode	C709	C710	R719	R701	R702	R703	C714	C715	ÊÇ·ñÖ§³Ö¶à°´¼ü
ACC	WBf°1uF NBf°100nF	WBf°1uF NBf°100nF	NC	1K	1.5K	1K	1uF	4.7uF	Y
DCC	0R	0R	NC	NC	NC	NC	NC	0R	N
DCC	0R	0R	2.5K	NC	NC	NC	NC	0R	Y

MIC -- Sub



EAR_JACK



Earphone Detection Circuit				
Mode	R725	R724	R734	R720
Low Cost Mode	NC	NC	47K	NC
Traditional Mode (ACC)	470K	47K	NC	0R

COMPANY: TRANSSION			
TITLE: INF-NOTE4 LITE_MT6737T_Schematic			
DRAWN: Longbaicheng	DATED: 2019-01-13		
CHECKED: <Checked By>	DATED: <Checked Date>		
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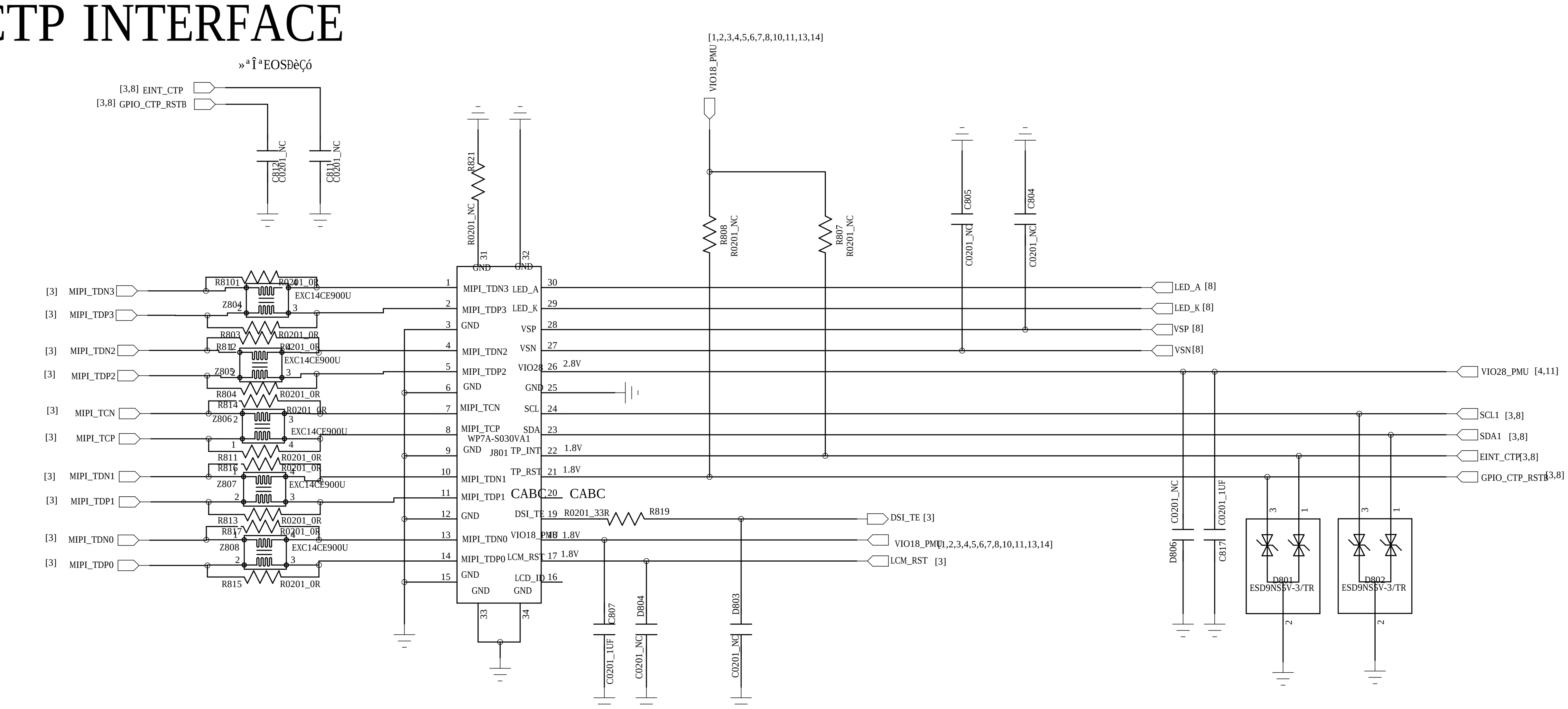
SCALE: <Scale>	SHEET: 7 OF 17		

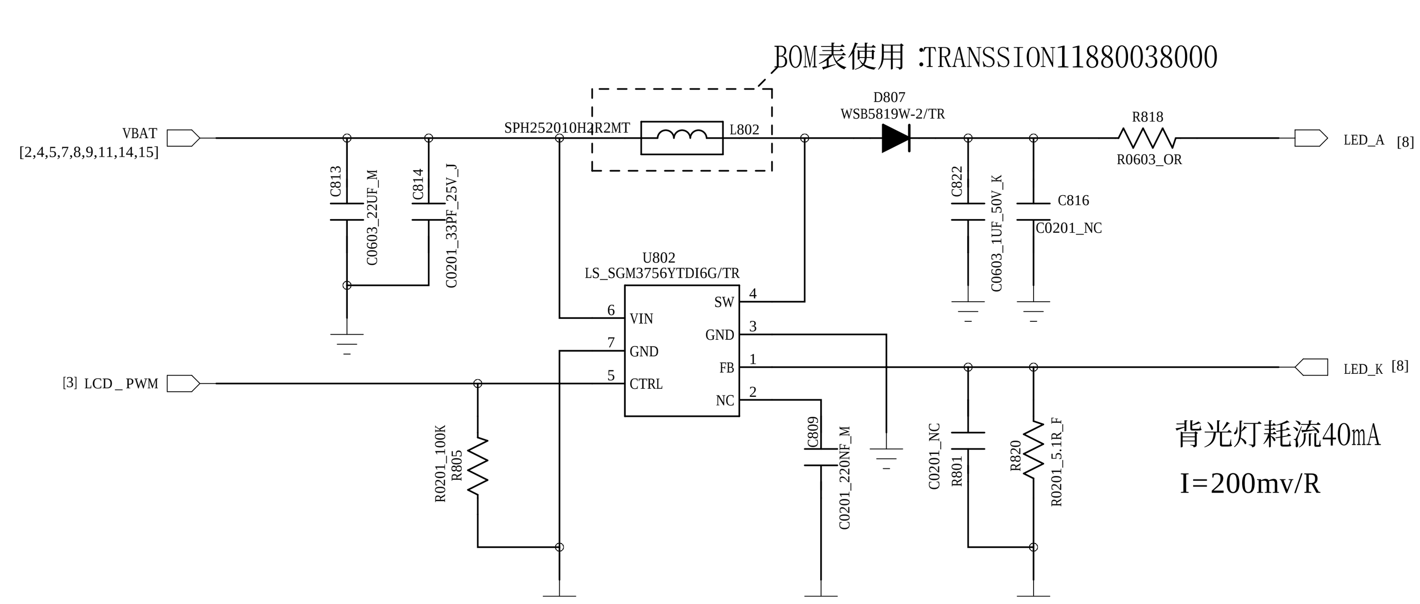
LCM ÐÂ¿ª

LCD AND CTP INTERFACE

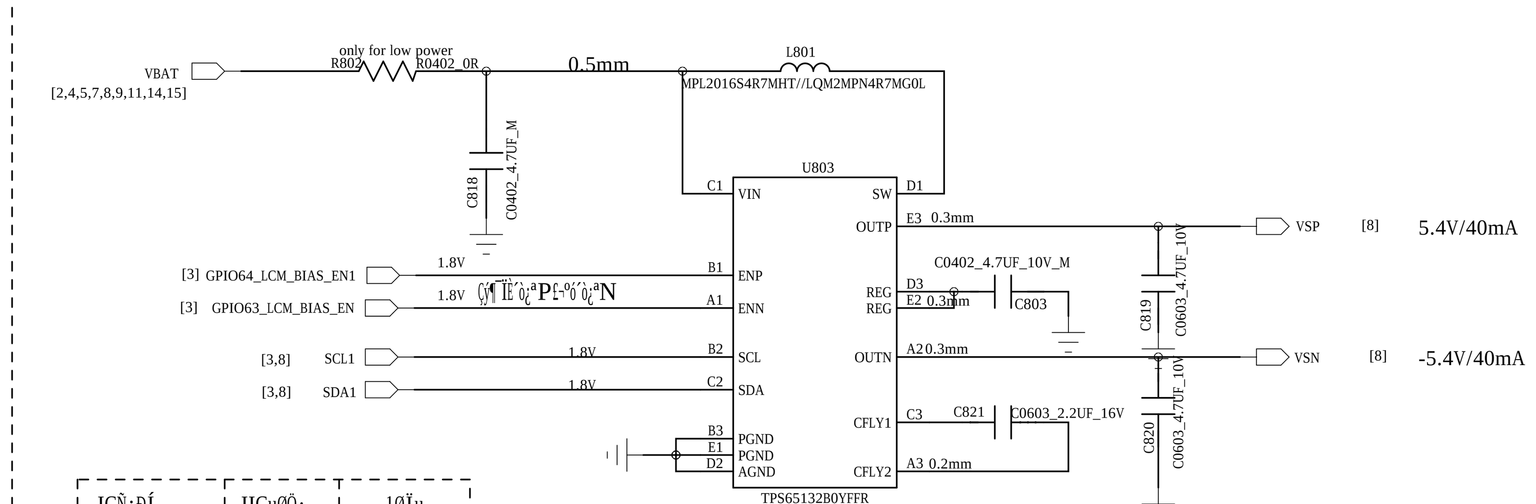


LCM Backlight LED Driver

from AL1518



LCD +/-5V Driver



ICÑ¡ÐÍ	IICµØÖ·	¹ØÏµ
SM5106	0X3E	Ö÷¹©
NT50358CG/J	0X3E	¶þ¹©

COMPANY:	TRANSSION		
TITLE:	INF-NOTE4 LITE_MT6737T_Schematic		
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RELEASED: <Released By>	DATED: NA	SCALE: <Scale>	SHEET: 8 OF 17


Main CAMERA
8M¼æÈÝ13M
VCAMD(0.9/1.0/1.1/1.22/1.3/1.5) Ã»ÓÐ1.8V
Sub Camera
2M¼æÈÝ5M
24pinÁ¬½Ó£¬Ê¹ÓÃAL812µÄ
ºóÉãAVDD¼æÈÝ13MÍâÖÃLDO
U903
BU28TD2WNVX
Ç°Éã²¹¹âµÆ
D913
LS_A-SL680AW1D-HA3-2T
U905
LS_WD3100B-6/TR
WD3100£º
ILED=4.8K/RSET£¨78mA£©
EN¶ËÊäµÄÃ¬ÈÏÊ¹ÓÃ600Hz
NC
Q903
2SC4617TLQ
Main Flash LED Driver
BOM_Ã³ÉSGM3785
U901
SPH252010H1R0MT
1uH
ZC £º×ö³ÉÊ¹ÓÃSPH252010L1R0MT
PWM
I(flash)=8556/RSETF*2
I(MOVIE)=7063/RSETM*2
TRANSSION
INF-NOTE4 LITE_MT6737T_Schematic
DRAWN: Longbaicheng
DATED: 2019-01-13
CHECKED: <Checked By>
DATED: <Checked Date>
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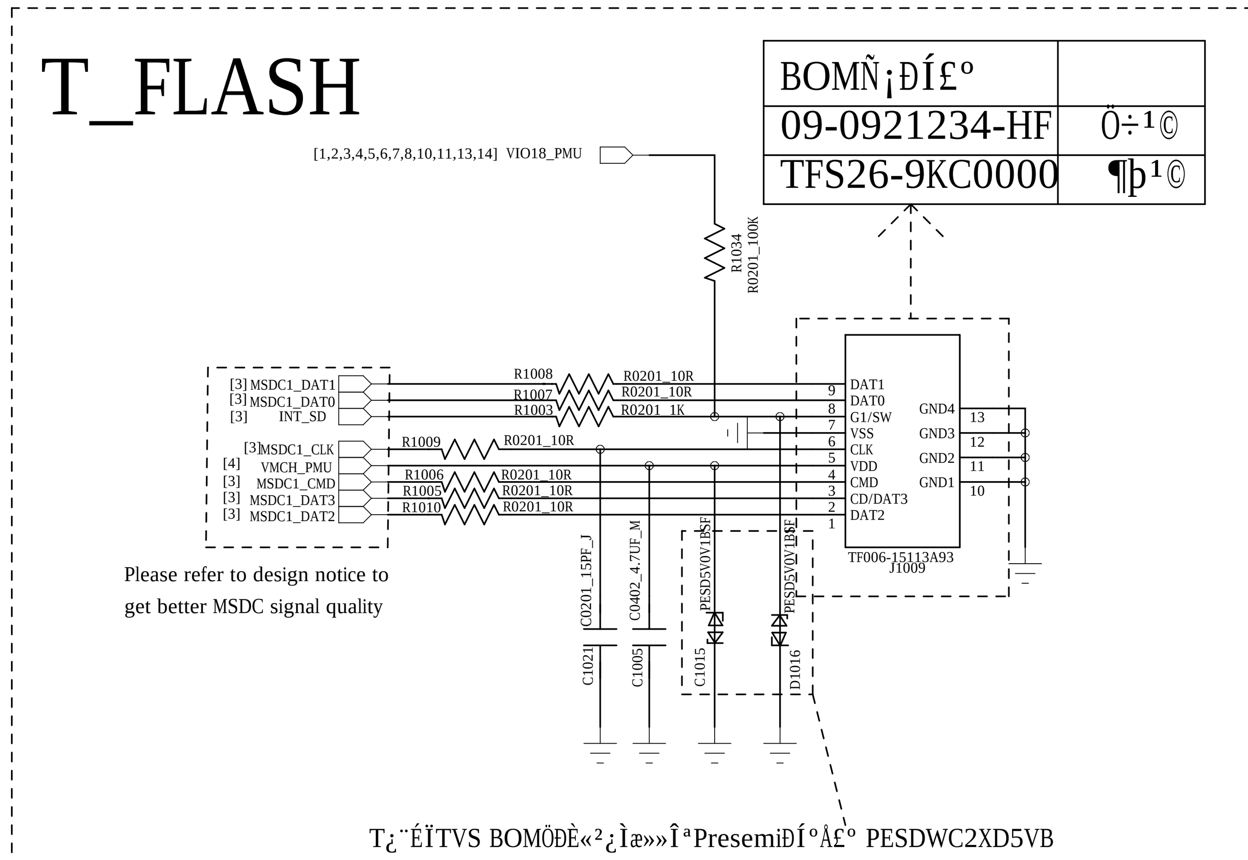
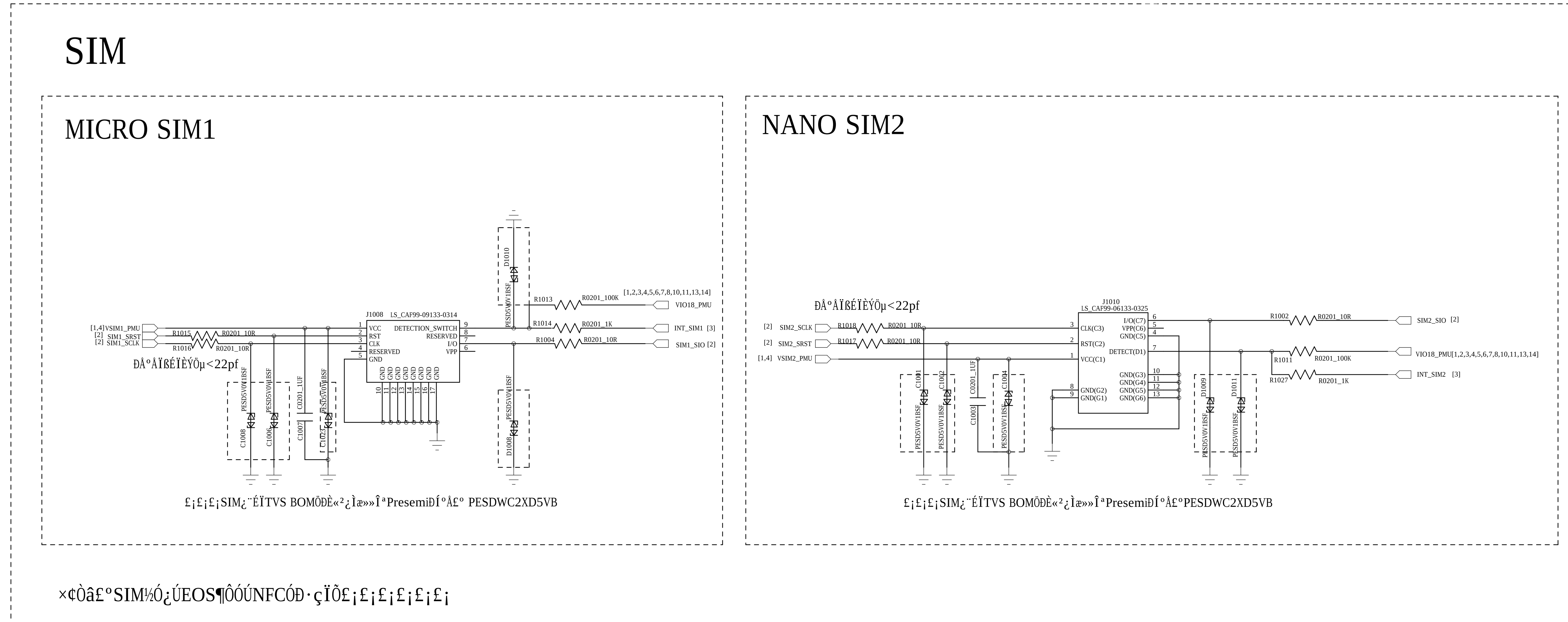
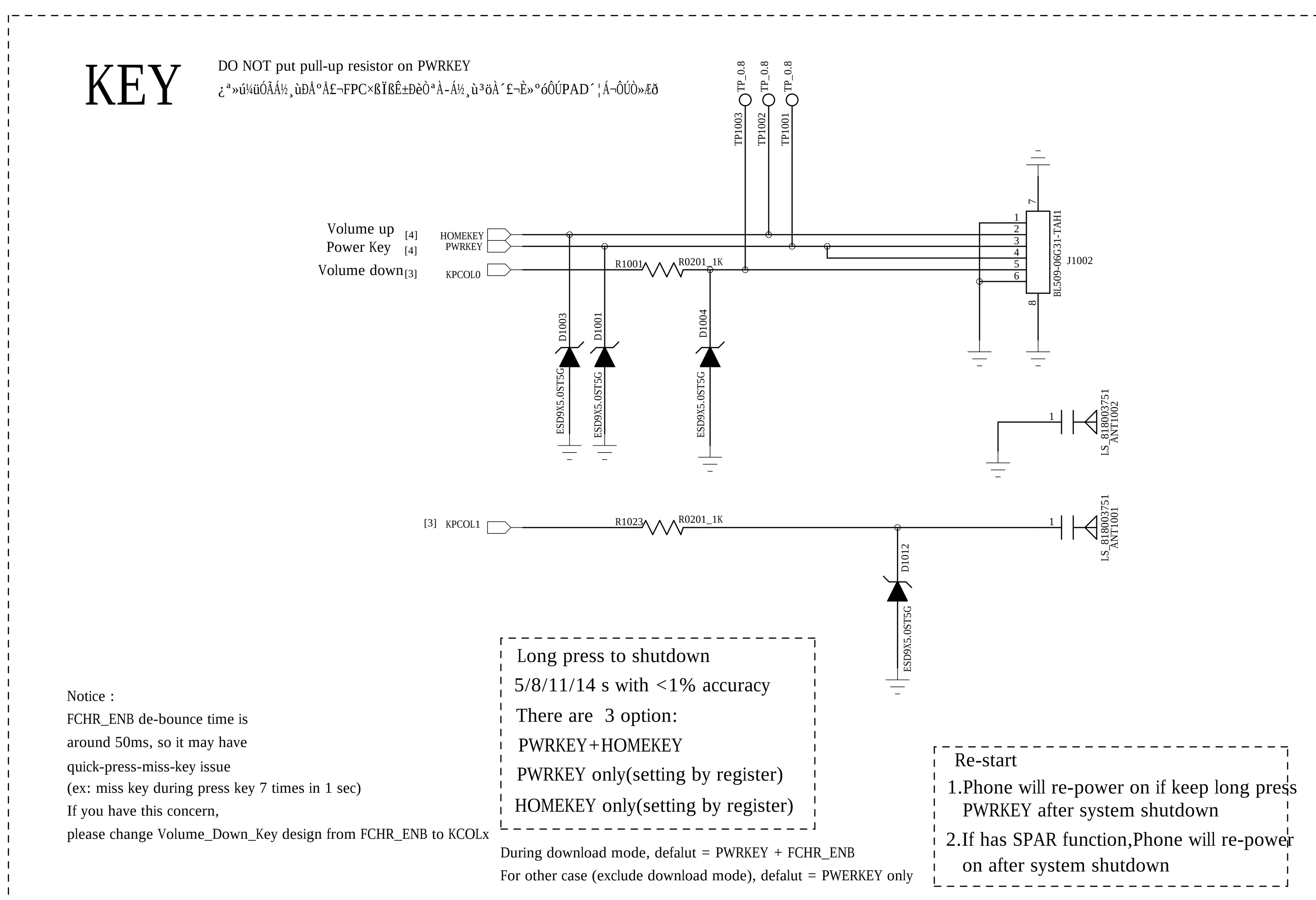


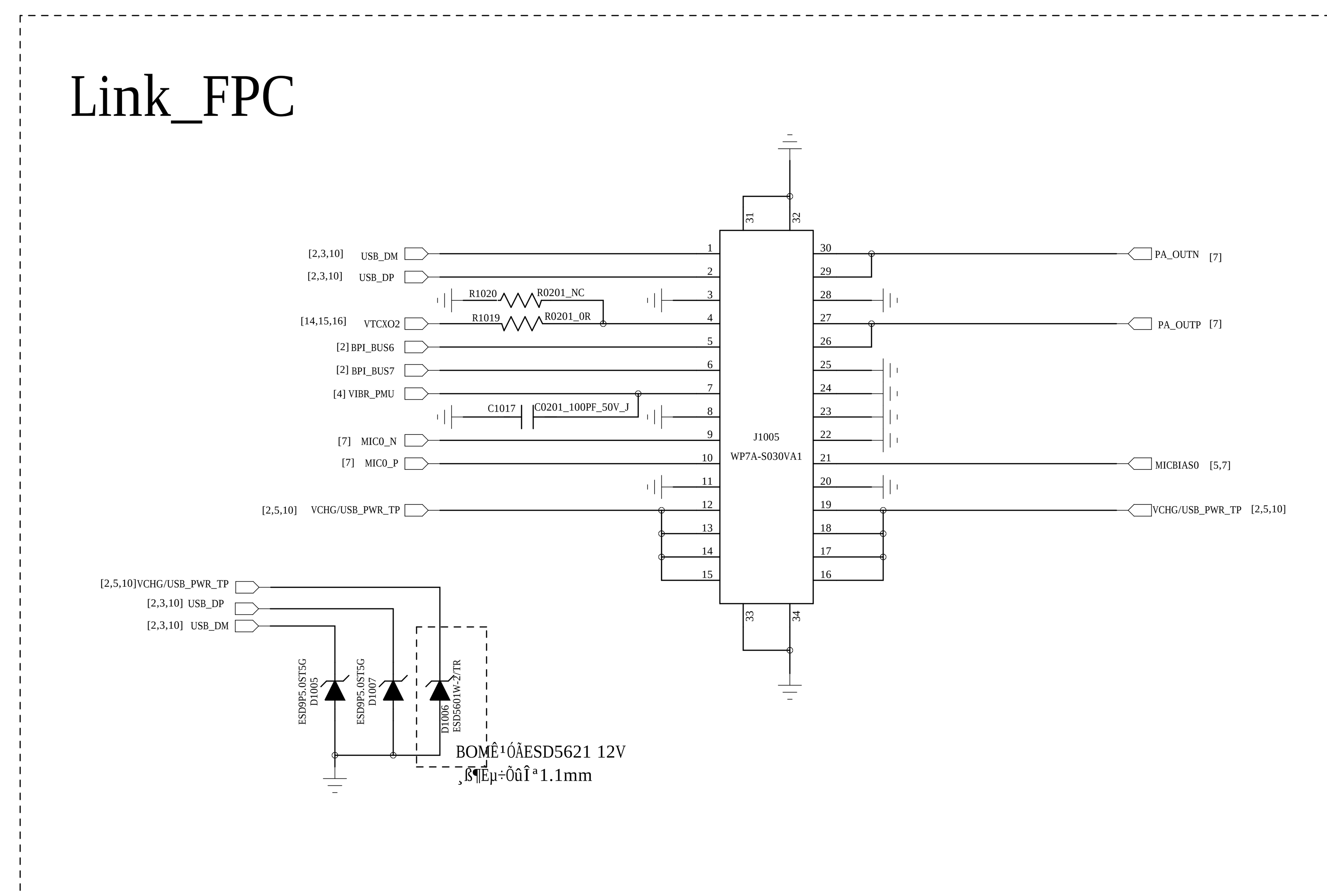


Schematic design notice of "40_MEMORY_SD Card" page.

Note 40-1:

The total equivalent capacitance of MSDC ESD protection device and de-sense bypass CAP must be <=10pF.
But for NFC app. equivalent capacitance of MSDC_NFC_SWPIO and MSDC_NFC_VCCSWP should <=0.5pF.





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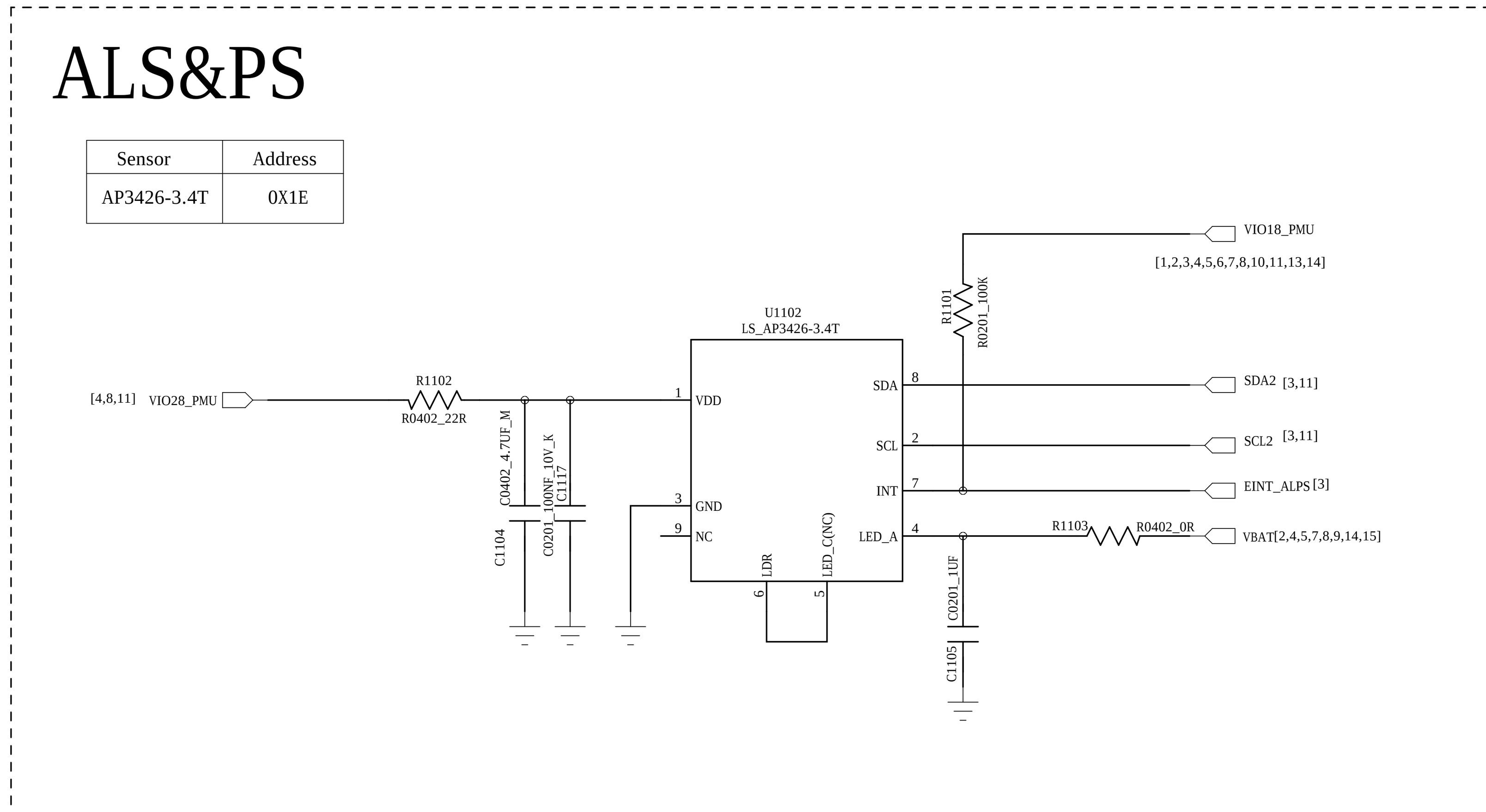

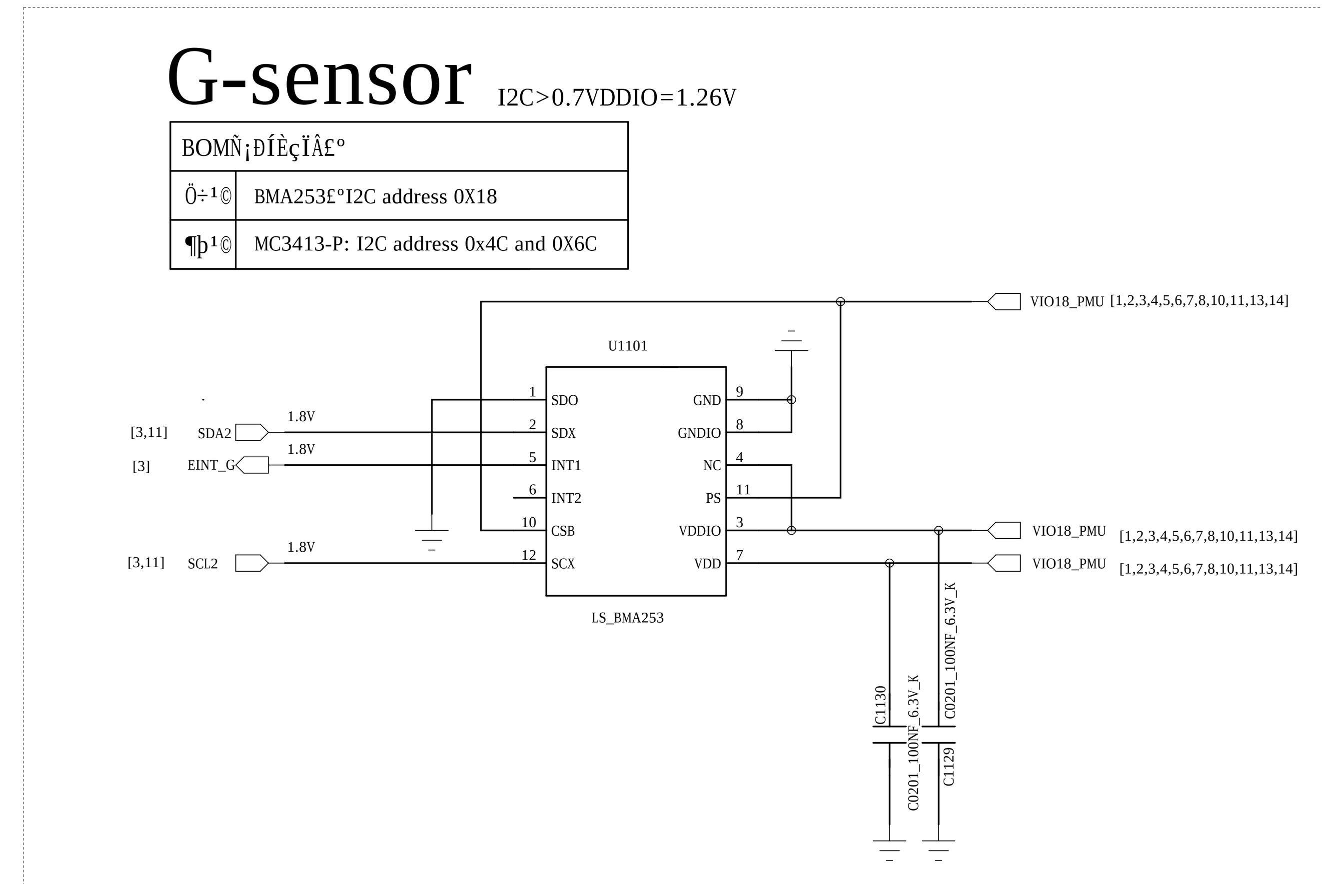

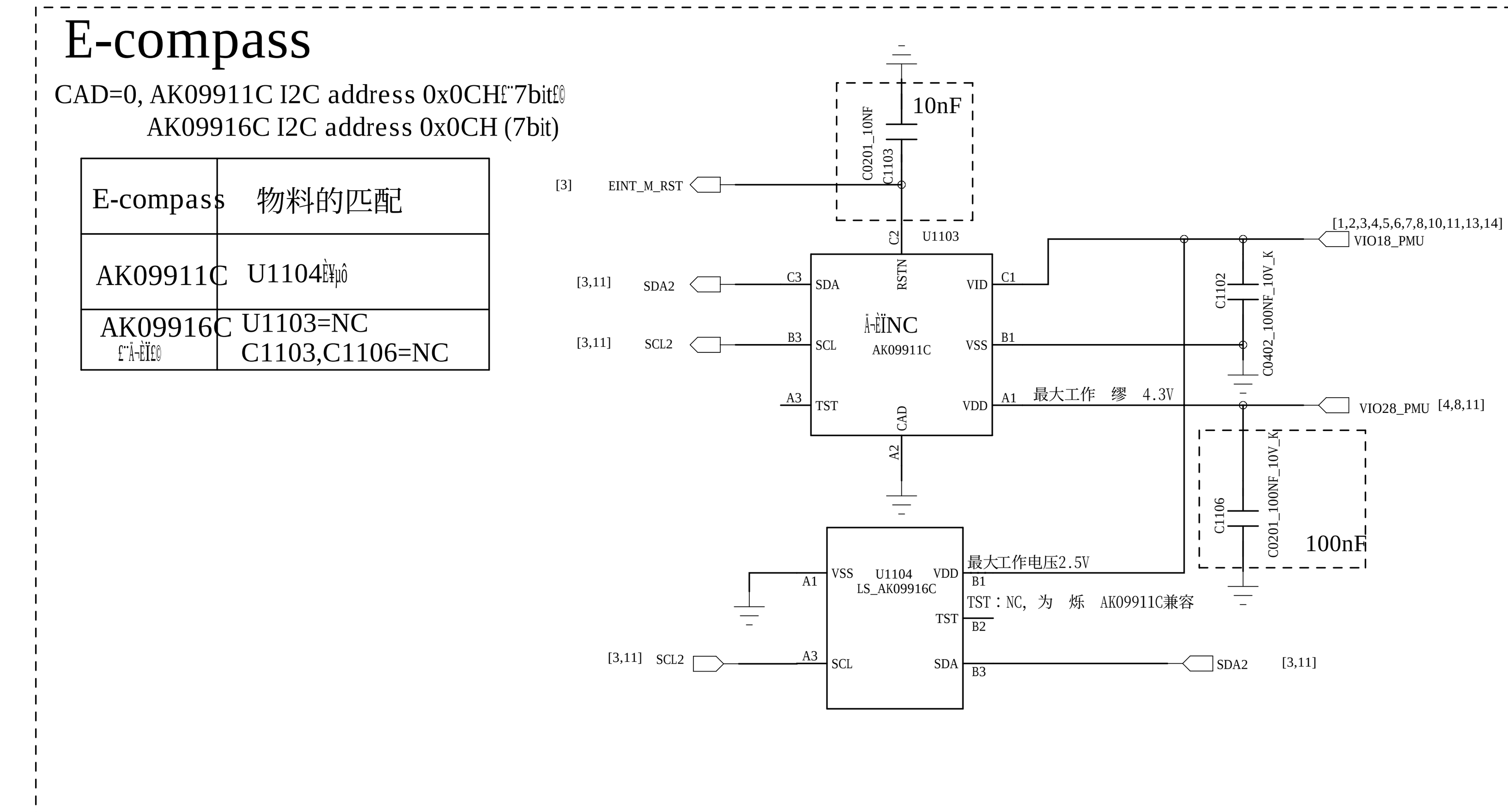

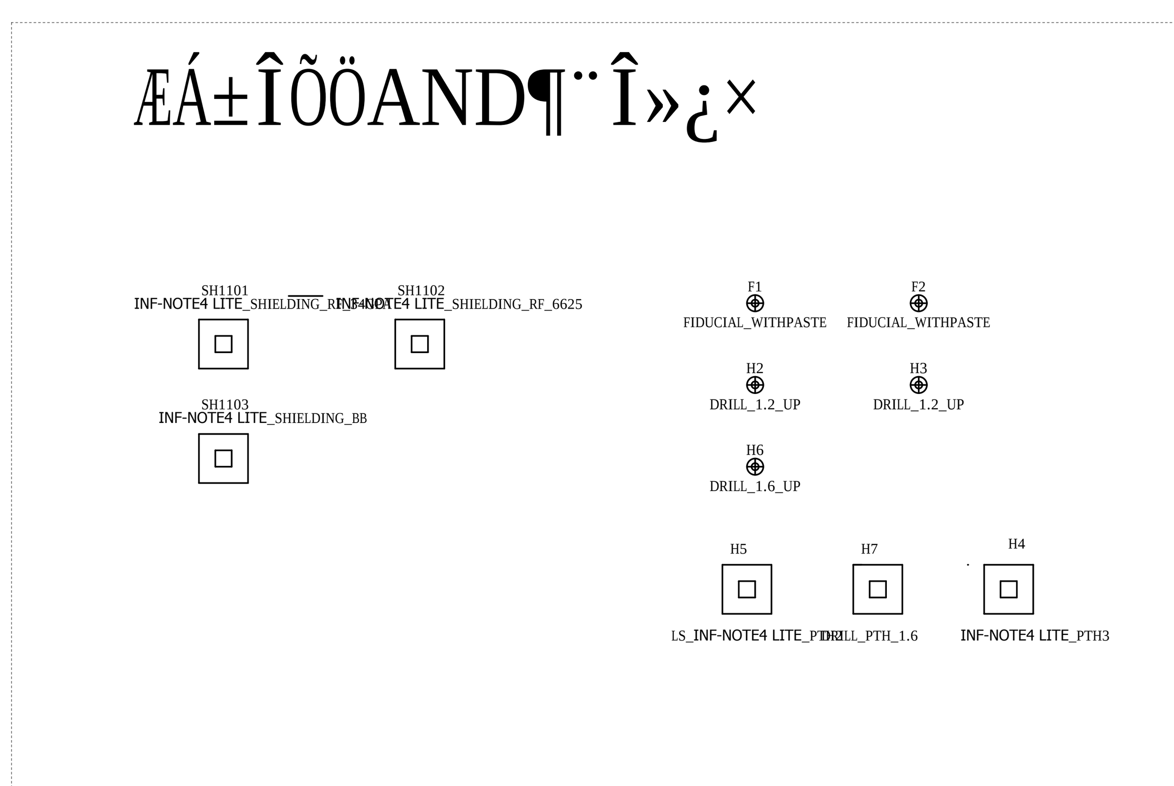

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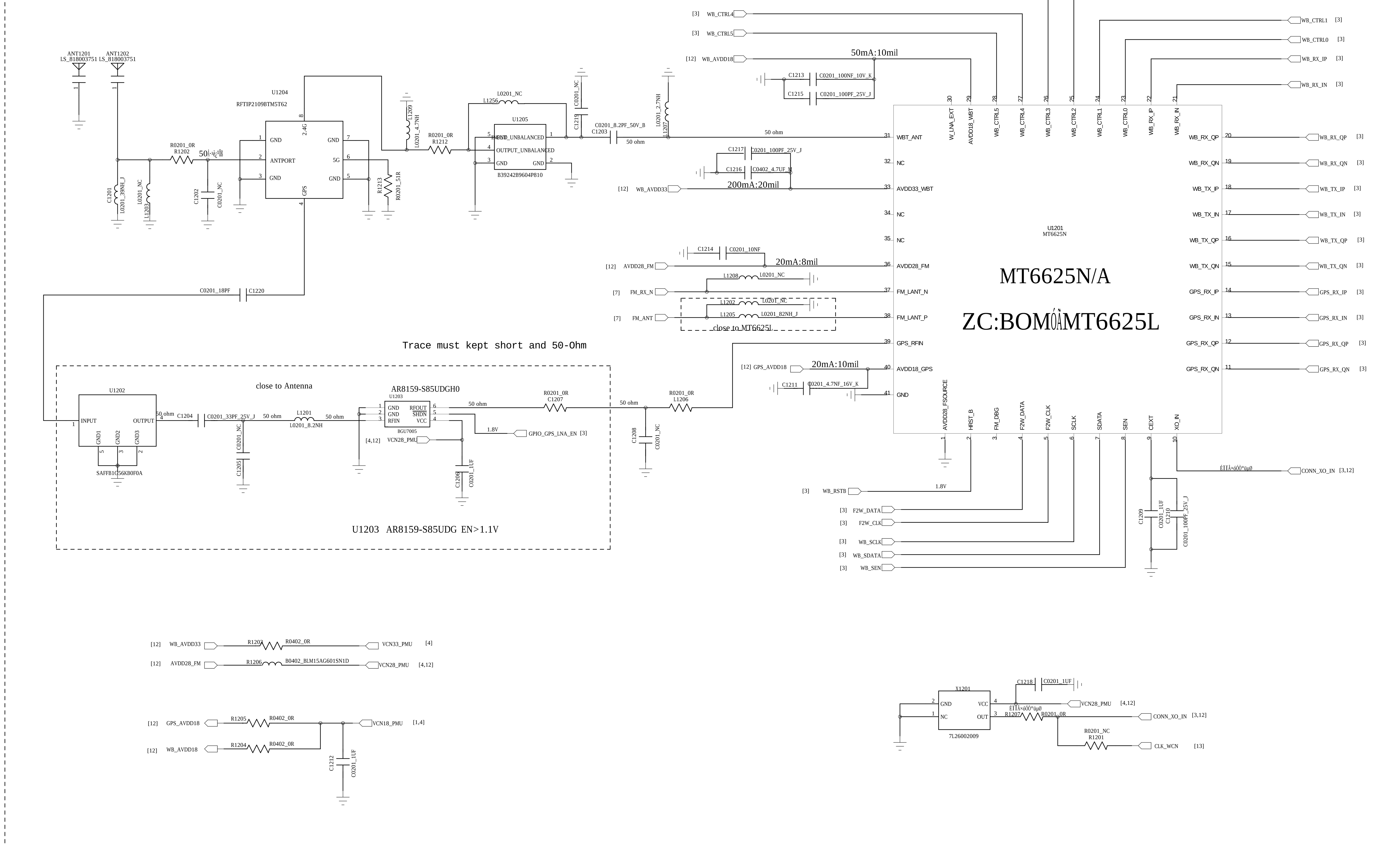
LTR	ECO NO:	APPROVED:	DATE:

GPS/WIFI/BT/FM



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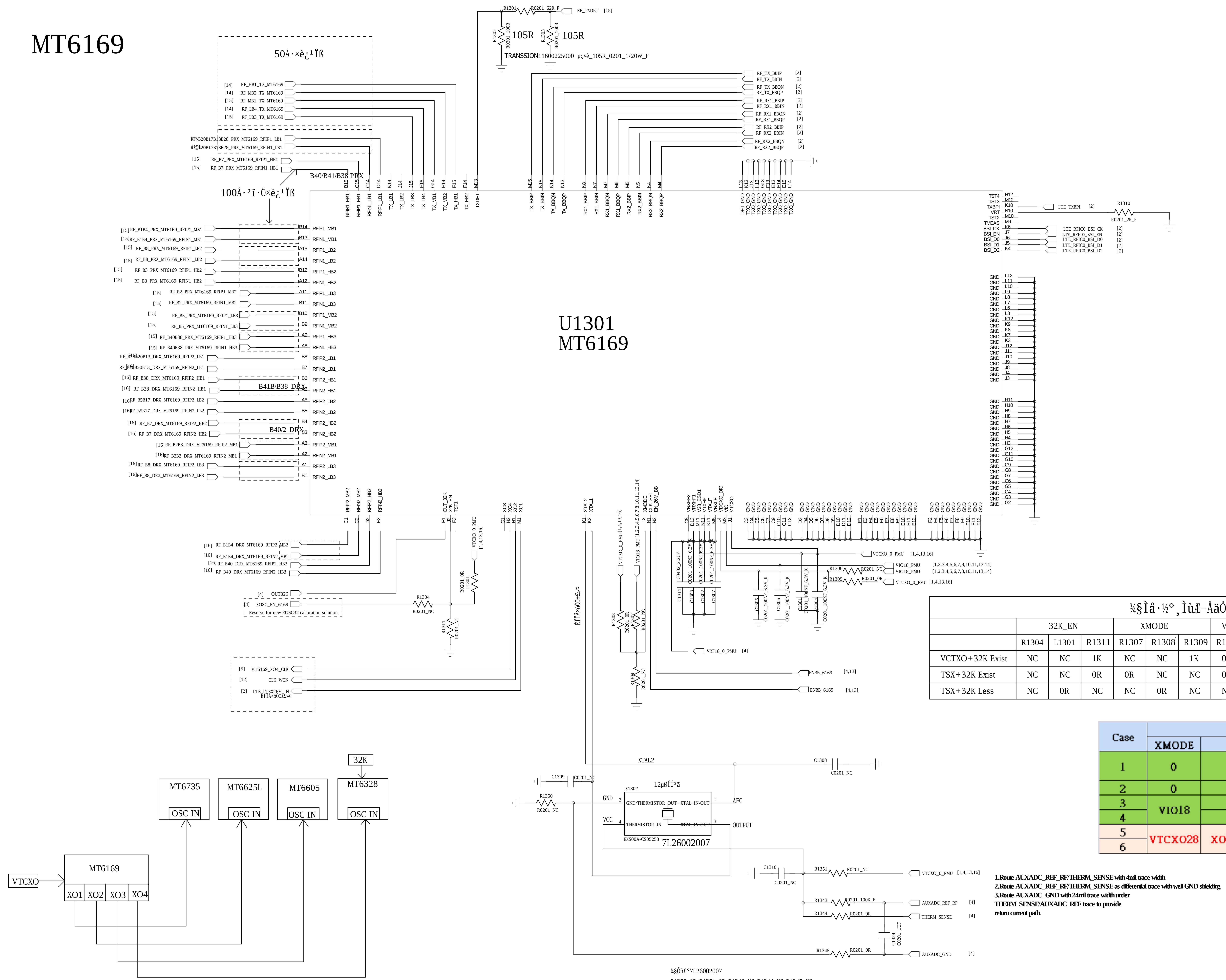
RELEASED: <Released By>	DATED: NA	SCALE: <Scale>		SHEET: 12 OF 17	

MT6169



¾§Ìå·½°¸ÌùÆ¬ÅäÖÃ±í2/3 (Page4¡¢13¡¢16£©																	
	32K_EN			XMODE			VDCXO_DIG		VCTCXO/TSX Component								
	R1304	L1301	R1311	R1307	R1308	R1309	R1306	R1305	C1308	C1309	R1350	R1351	R1343	R1344	R1345	C1310	C1324	X1302
VCTXO+32K Exist	NC	NC	1K	NC	NC	1K	0R	NC	100nF	NC	0R	0R	NC	NC	NC	100nF	NC	VCTXO
TSX+32K Exist	NC	NC	0R	0R	NC	NC	0R	NC	NC	NC	NC	NC	100K	0R	0R	NC	1uF	TSX
TSX+32K Less	NC	0R	NC	NC	0R	NC	NC	0R	NC	NC	NC	NC	100K	0R	0R	NC	1uF	TSX

Case	MT6169 Control Pin					MT6169 CLK Output			Clock scheme
	XMODE	32K_EN	VDXCO_DIG	EN_26M_BB	CLK_SEL	OUT_32K	XO1	XO2~4	
1	0	0	VIO18	0	0	Off	Off	Off	VCTXO 32K XO
2	0	0		1	1	Off	26M Out	26M Out	
3	VIO18	0		0	0	Off	Off	Off	26M XO
4		0		1	1	Off	26M Out	26M Out	32K XO
5	VTCXO28	XOSC_EN_6169	VTCXO28	0	0	On	Off(LPM)	Off	26M XO
6				1	1	On	26M Out	26M Out	32K Less

1.Route AUXADC_REF_RF/THERM_SENSE with 4mil trace width
2.Route AUXADC_REF_RF/THERM_SENSE as differential trace with well GND shielding
3.Route AUXADC_GND with 24mil trace width under THERM_SENSE/AUXADC_REF trace to provide return current path.

¾§Õñ£º7L26002007
R1350=0R,R1351=0R,R1343=NC,R1344=NC,R1345=NC
C1310=100NF,C1324=NC,C1308=100NF,C1309=NC

¾§Ìå£ºCT2520DB26000C0FZZA1£¨MTK½¨ÒéÓÃ²Î¿¼Í¼ÉÏµÄ£¬ÔÝÎÞÑéÖ¤ÁÐ±í£©
R1350=NC,R1351=NC,R1343=100K,R1344=0R,R1345=0R
C1310=NC,C1324=1UF,C1308=NC,C1309=NC

26M¾§Õñ£ºC430=0R£»26M¾§Ìå£º1UF

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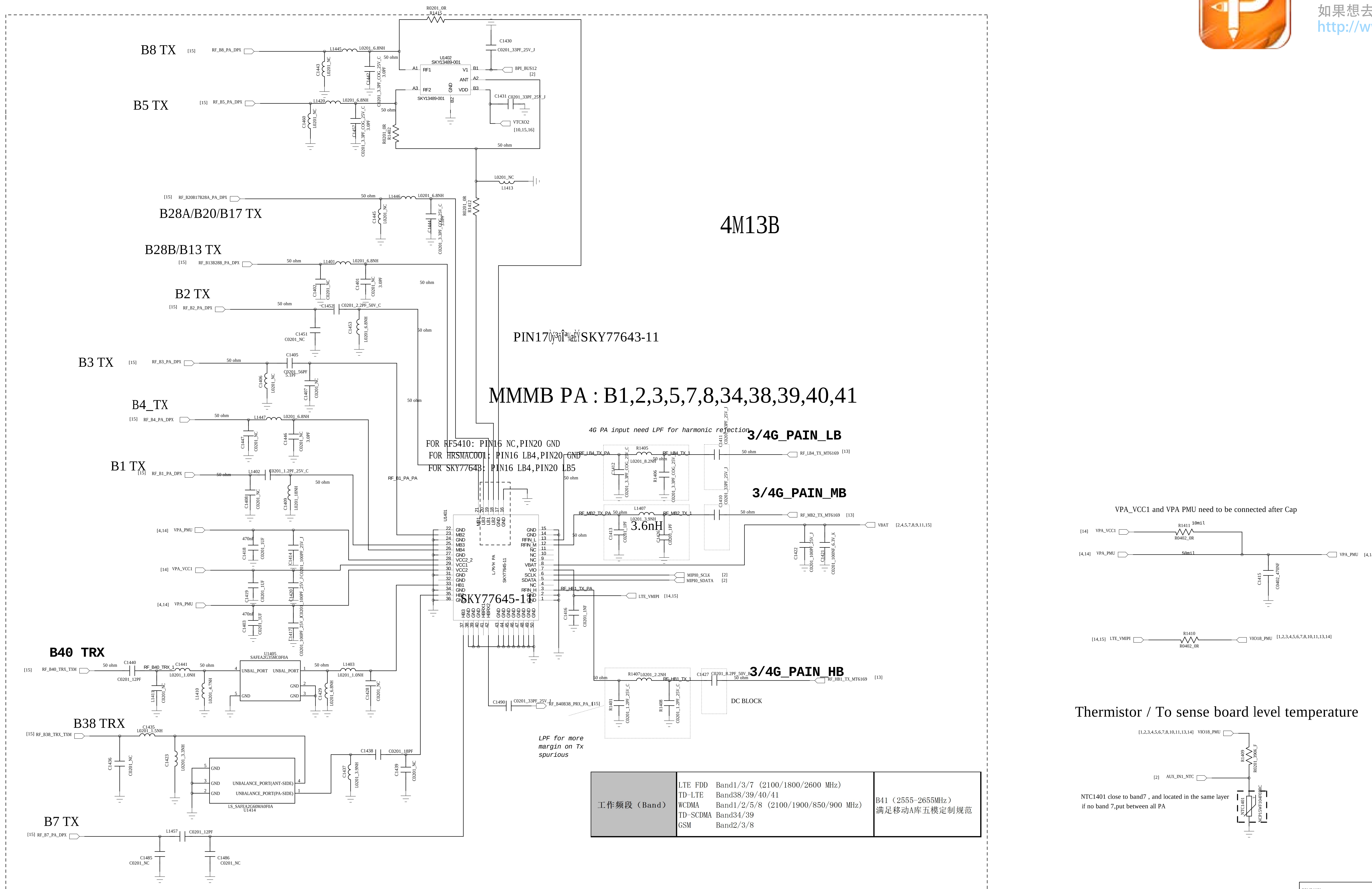

工作频段（Band）			
	LTE FDD	Band1/3/7 (2100/1800/2600 MHz)	B41 (2555-2655MHz) 满足移动A库五模定制规范
	TD-LTE	Band38/39/40/41	
	WCDMA	Band1/2/5/8 (2100/1900/850/900 MHz)	
	TD-SCDMA	Band34/39	
	GSM	Band2/3/8	

COMPANY: TRANSSION

TITLE: INF-NOTE4 LITE_MT6737T_Schematic

DRAWN: Longbaicheng	DATED: 2019-01-13	CODE: <Code>	SIZE: D
CHECKED: <Checked By>	DATED: <Checked Date>	DRAWING NO: <Drawing Number>	REV: V1.0
QUALITY CONTROL: <QC By>	DATED: <QC Date>		
RELEASED: <Released By>	DATED: NA	SCALE: <Scale>	SHEET: 14 OF 17


ASM_Main
77916:24~37,77912:27~36,77910:28~35
SP12T TXM 2G/TD
SKY77912-51
NC£¬RF PA EOS
B2 TRX
B8 TRX(Áª Í¨ÅäÖÃ)
SAYEY897MHA0F0A
100Å·²î·Ö×ß¿¹Îß
MT6169 TX Ouput need a DC block(MUST).
2G_PAIN_HB
High pass filter for G8 Tx in G7 Rx band noise rejection
2G_PAIN_LB
B3 TRX
B5 TRX
B7 TRX
BOM ¸Ä SAYEY2G53CA0F0A
B40/B38 PRX
B1 TRX
B4 TRX
SKY13489-001
VTCXO2
BPI_BUS10
RF_TXDET
VBAT
LTE_VMIPI
MIPI1_SDATA
MIPI1_SCLK
TRANSION
INF-NOTE4 LITE_MT6737T_Schematic
DRAWN: Longbaicheng
DATED: 2019-01-13
CHECKED: <Checked By>
DATED: <Checked Date>
QUALITY CONTROL: <QC By>
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DATED: NA
CODE: <Code>
SIZE: D
DRAWING NO: <Drawing Number>
REV: V1.0
SCALE: <Scale>
SHEET: 15 OF 17

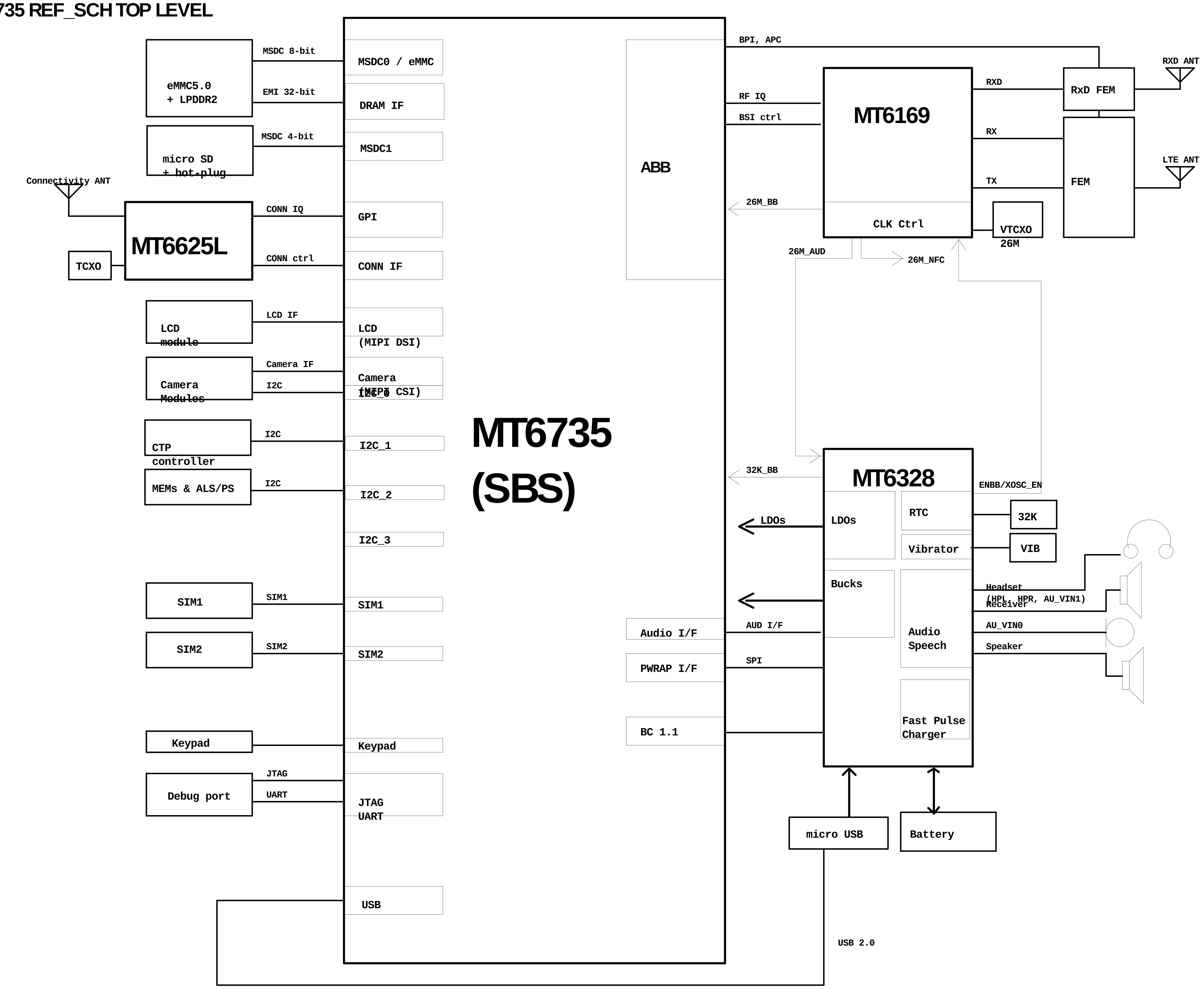
¼§Ìå·½º¸Ìù£¬ÅäÖÃ±í3/3 (Page4¡¢13¡¢16£©		
	VTCXO2 Power	
	R1606	R1607
VCTXO+32K Exist	NC	0R
TSX+32K Exist	NC	0R
TSX+32K Less	0R	NC

COMPANY: TRANSSION

TITLE: INF-NOTE4 LITE_MT6737T_Schematic

DRAWN: Longbaicheng	DATED: 2019-01-13	CODE: <Code>	SIZE: D
CHECKED: <Checked By>	DATED: <Checked Date>	DRAWING NO: <Drawing Number>	REV: V1.0
QUALITY CONTROL: <QC By>	DATED: <QC Date>	SCALE: <Scale>	SHEET: 16 OF 17
RELEASED: <Released By>	DATED: NA		

Project : MT6735 REF_SCH TOP LEVEL



COMPANY:	TRANSSION		
TITLE:	INF-NOTE4 LITE_MT6737T_Schematic		
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CHECKED: <Checked By>	DATED: <Checked Date>	DRAWING NO:	REV: V1.0
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